# Reproducing conversational recall in SimpleMem

1. Project and reproduction target

For this project, I reproduced the project's LoCoMo question-answering workflow and tested it on 20 fixed questions. In order to find how sensitive are its answers to the number of memories retrieved by semantic search. And this reproducibility has reduced the semantic retrieval limit from 25 to 10 raised mean token F1 from 0.4528 to 0.4687 across three trials per setting. The advantage reversed in the third trial, and mean retrieval time differed by less than 0.02 seconds per question.

SimpleMem is an agentic memory system. It turns conversation history into searchable memories, retrieves evidence in response to a question, and uses a language model to generate an answer from that evidence. Running the authors' evaluation script exercises this sequence, including memory construction and retrieval. My reproduction target was the demonstrated ability to answer questions from stored conversation history. The modification tested whether a smaller semantic candidate set would preserve answer quality and reduce retrieval time.

2. Implementation and data

| **Component** | **Setting used in this study** |
|---|---|
| Code and environment | SimpleMem commit db80b6a7c591e0ea730a058e9f5fc 4eb06572299, Google Colab, Python 3.13, OpenAI Python client 2.3.0 |
| Language model | DeepSeek API, requested model deepseek-v4-flash, base URL https://api.deepseek.com. The connectivity test returned the model label deepseek-flash. |
| Model adaptation | The repository template specifies gpt-4.1-mini. Both experimental conditions used DeepSeek through the same OpenAI-compatible interface. The API key was stored in Colab Secrets. |
| Embeddings | Qwen/Qwen3-Embedding-0.6B with 1,024-dimensional vectors |
| Evaluation data | LoCoMo conversation conv-26. A fixed selection of 20 of its 199 questions used sampling seed 42. Each run retained the conversation history. |
| Experimental variable | SEMANTIC_TOP_K was 25 in the local baseline and 10 in the modification. KEYWORD_TOP_K and STRUCTURED_TOP_K were both 5. |

I used the same question file in all six evaluations. The subset contained seven questions from category 1, six from category 2, one from category 3, five from category 4 and one from category 5. Seed 42 fixes question selection. Repeated runs measure variability on those same questions. Hardware type, effective decoding settings (temperature, top-p and maximum

output tokens), and any provider-side defaults were not recorded in the run summaries. No prompt change was part of the experimental design.

## 3. Evaluation and results

I ran the root-level test_locomo10.py script once per trial with the subset as its dataset, one conversation sample and a distinct result file. I used three measures: token F1, the harmonic mean of token precision and recall against the reference answer; exact match, the fraction of answers the evaluator scored as a complete match; and retrieval time, the script's mean retrieval-stage time per question. This timing excludes answer generation and does not measure total notebook runtime.

| Trial | Top-25 F1 | Top-10 F1 | Top-25 matches | Top-10 matches | Top-25 s/question | Top-10 s/question |
|---|---|---|---|---|---|---|
| 1 | 0.4175 | 0.4806 | 4/20 | 5/20 | 24.63 | 24.12 |
| 2 | 0.4335 | 0.4829 | 4/20 | 5/20 | 22.17 | 20.14 |
| 3 | 0.5072 | 0.4425 | 5/20 | 4/20 | 20.11 | 22.59 |
| Mean | 0.4528 | 0.4687 | 21.67% | 23.33% | 22.30 | 22.28 |

The change in mean F1 was +0.0159, or 1.59 points on a 100-point scale. Across 60 responses per setting, Top-10 produced 14 exact matches and Top-25 produced 13. These totals represent repeated answers to 20 distinct questions. The sample variance of the three trial-level F1 scores was 0.002290 for Top-25 and 0.000516 for Top-10. The corresponding sample variances were 0.000833 for both exact-match proportions and 5.1234 versus 4.0136 for mean retrieval time in seconds squared. With only three trials, the variance estimates are sensitive to individual runs.

Comparison with the original project. The authors' project page reports 43.24% F1 on LoCoMo. My local baseline averaged 45.28% on one conversation subset using DeepSeek. The model, question mix and evaluation scope differ. Consequently, the numerical proximity does not establish replication of the published benchmark score. The direct comparison supported by this experiment is between the two retrieval settings on identical questions.

## 4. What changed at the question level

Temporal wording. In Trial 3, the question "When did Caroline give a speech at a school?" had the reference "The week before 9 June 2023." Top-25 answered "In the week before 9 June 2023" (F1 0.923). Top-10 answered "During the week of 29 May 2023" (F1 0.462). Top-25 retained the reference's time anchor. Top-10 introduced a calendar interpretation and shared fewer tokens with the reference. The score alone cannot settle whether the underlying conversation supports that interpretation.

Missing the requested fact. For "What is Caroline's relationship status?", the reference was "Single." Top-10 said the status was unspecified. Top-25 mentioned a past breakup but also said the current status was unspecified. Both received F1 0. The larger candidate limit

produced a more detailed answer in this run, yet the extra detail did not resolve the question. Determining whether the relevant fact was absent from retrieval or overlooked during generation would require inspecting the retrieved passages.

Across all 20 questions in Trial 3, Top-25 had higher F1 on eight and tied on twelve. Its mean F1 advantage that round was 0.0648. This reversal qualifies the small three-trial advantage for Top-10 and explains why a single run would give an incomplete account.

5. Debug diary and reproduction notes

| **Symptom** | **Resolution or consequence** |
|---|---|
| TensorFlow import failed because Protobuf-generated code used version 6.31.1 and the runtime used 5.29.5. | I used the PyTorch path and set USE_TF=0 and USE_TORCH=1 before launching the evaluation subprocess. |
| Evaluation reached BLEU scoring and stopped because the NLTK punkt_tab resource was missing. | I downloaded punkt_tab and checked word_tokenize before rerunning. The subsequent one-question test completed and wrote its JSON output. |
| The DeepSeek connectivity check returned HTTP 402 with “Insufficient Balance.” | The error identified an API-account balance issue. A successful model request was a prerequisite for the evaluation runs. |

As an end-to-end check, the system answered the support-group question with “07 May 2023” against the reference “7 May 2023.” It retrieved 31 items, produced the correct calendar date and received exact match 0 and token F1 0.667. This case shows both that the pipeline recovered a conversational fact and that surface formatting can affect the score. The 31 items also illustrate why a semantic branch limit should not be read as the total number of items returned across retrieval channels.

6. Conclusion and replication details

The released pipeline reproduced conversational recall and generated question-level scores under an alternative model provider. On this fixed subset, lowering the semantic retrieval limit gave a modest average F1 gain and almost identical mean retrieval time. Performance depended on the trial. One conversation offers limited evidence about other speakers, question categories or the full LoCoMo benchmark. A follow-up should evaluate more conversations and retain the retrieved passages for questions whose answers change.

To repeat the comparison, check out the commit listed above, install its root requirements, configure the model and embeddings, and use the archived locomo_conv26_seed42_20q.json. From the repository root, run the following command for each setting and trial, changing the result filename each time. The argument –num-samples 1 selects the one conversation containing all 20 questions.